

Giải thuật Kruskal

```
1 #include <stdio.h>
2 #define MAX_N 100
3 typedef struct {
4     int u, v, w;
5 }Edge;
6
7 typedef struct {
8     Edge edges[MAX_N];
9     int n, m;
10 }Graph;
11
12 void init_graph(Graph *G, int n) {
13     G->n=n;
14     G->m=0;
15 }
16
17 void add_edge(Graph *G, int u, int v,int w) {
18     G->edges[G->m].u = u;
19     G->edges[G->m].v = v;
20     G->edges[G->m].w = w;
21     G->m++;
22 }
23
24 int parent[MAX_N];
25
26 int findRoot(int u) {
27     if (parent[u] == u)
28         return u;
29     else return findRoot(parent[u]);
30 }
31
32 int Kruskal(Graph *G, Graph *T) {
33     int u, v, w, e;
34     for (u=0;u<G->m;u++)
35         for (v=u+1;v<G->m;v++)
36             if (G->edges[u].w > G->edges[v].w) {
37                 Edge t = G->edges[u];
38                 G->edges[u]= G->edges[v];
39                 G->edges[v] = t;
40             }
41
42     init_graph(T, G->n);
43     for (u=1;u<=G->n;u++)
44         parent[u] = u;
45     int sum_w = 0;
46
47     for (0;e<G->m;e++) {
48         u = G->edges[e].u;
49         v = G->edges[e].v;
50         w = G->edges[e].w;
51
52         int root_u = findRoot(u), root_v = findRoot(v);
53
54         if (root_u!=root_v) {
55             add_edge(T, u, v, w);
56
57             parent[root_v] = root_u;
58             sum_w += w;
59         }
60     }
61
62     return sum_w;
63 }
```

```

65 int main() {
66     Graph G, T;
67     int n, m, u, v, w, e;
68     FILE *f = fopen("Kruscal_dt_130.txt", "r");
69     // FILE *f = fopen("Kruscal_dt_163.txt", "r");
70     fscanf(f, "%d%d", &n, &m);
71     init_graph(&G, n);
72
73     for (e=1;e<=m;e++) {
74         fscanf(f, "%d%d%d", &u, &v, &w);
75         add_edge(&G, u, v, w);
76     }
77
78     int sum = Kruskal(&G, &T);
79     printf("Cay khung co trong so nho nhat la %d", sum);
80     for (e=0;e<T.m;e++)
81         printf("\n%d %d %d", T.edges[e].u, T.edges[e].v, T.edges[e].w);
82
83     fclose(f);
84     return 0;
85 }

```